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RESEARCH ARTICLE

Label-Free Biomolecule Detection With InP/AlGaAs Charge Plasma Dielectric-Modulated Vertical TFET Using TaN as Metal Gate

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ABSTRACT This study introduces a label-free biosensing method for biomolecule detection utilizing an InP/AlGaAs charge plasma dielectric-modulated vertical tunnel field-effect transistor (InP/AlGaAs VTFET) featuring TaN as the metal gate. The device comprises a nanocavity beneath the gate metal adjacent to the tunnelling junction, where biomolecules engage with the dielectric material, resulting in fluctuations in the drain current. A twin metal gate architecture with laterally split dielectrics is employed to eliminate short-channel effects. The biosensor has exceptional sensitivity, attaining a peak drain current sensitivity of 2.5×10 cm^2 for biomolecules such as albumin ($k = 7$). The gadget proficiently identifies biomolecules with varying dielectric constants and charge distributions, enabling adaptable label-free detection of diverse targets. The study examines the influence of dielectric constant on critical metrics such as Energy Band Diagrams, Drain Current, Drain Sensitivity, and subthreshold swing (SS). The overlap between the source and pocket regions, along with the introduction of an auxiliary gate, optimizes the electrical characteristics. Simulation results show that the proposed InP/AlGaAs VTFET achieves a maximum sensitivity of 3.5×10^3, outperforming configurations without overlap (2×10^3), highlights the potential of proposed InP/AlGaAs VTFET for scalable, high-sensitivity, label-free biomolecule detection.

INDEX TERMS Biomolecule detection, dielectric-modulated TFET, drain current sensitivity, InP/AlGaAs VTFET, label-free biosensing.

I. INTRODUCTION

Label-free biosensors have attracted considerable interest in recent years because they can detect biomolecules without markers or identifiers, providing benefits such as lower costs, simplicity, and real-time analysis. Field-effect transistors (FETs) have emerged as a very promising platform for biosensing applications due to their capacity to detect subtle alterations in channel conductivity resulting from the binding of target biomolecules. However, traditional MOSFET-based biosensors encounter considerable obstacles, such as inadequate subthreshold swing ($SS > 60$ mV/dec), limited sensitivity, and excessive power dissipation resulting from thermionic electron emission and short-channel effects (SCEs), which constrain their efficacy in high-speed or low-power applications. A viable

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solution to these issues is the implementation of Tunnel Field-Effect Transistors (TFETs), which utilize band-to-band tunnelling (BTBT) for charge transport. The BTBT mechanism enables TFETs to surpass the physical constraints of MOSFETs, including the 60 mV/dec subthreshold swing, resulting in enhanced sensitivity, reduced power consumption, and accelerated switching speeds. Vertical Tunnel Field-Effect Transistors (VTFETs) exhibit significant promise for biosensing applications owing to their improved electrostatic control, less short-channel effects, and exceptional switching characteristics. The elevated sensitivity and minimal power dissipation attributes of TFETs render them particularly appealing for label-free biomolecule detection, essential for medical diagnostics, environmental monitoring, and biochemical applications. A primary problem in the development of highly sensitive TFET-based biosensors is the design of the device architecture. The efficacy of the biosensor can be substantially affected by variables like the gate dielectric material, the configuration of the nanogap, and the selection of metal contacts. Many researchers have investigated charge plasma manipulation and dielectric materials to improve the sensitivity of TFET-based biosensors. Modifying the dielectric constant (k) of the materials in the gate area enables precise calibration of the device's response to various biomolecules, ensuring elevated specificity and sensitivity for a diverse array of analytes. The application of heterostructures in TFETs has garnered interest owing to their enhanced electron mobility and adjustable bandgaps, which provide additional enhancements in biosensor performance [1], [2], [3], [4]. The integration of sophisticated gate materials, such as TaN, along with a dual metal gate architecture including laterally split dielectrics, has been suggested to enhance the electrical properties of the device and alleviate short-channel effects. These architectural innovations aim to augment the device's sensitivity, diminish ambipolar conduction, and raise the signal-to-noise ratio, so rendering the TFET-based biosensor more efficient for label-free biomolecule detection. Dielectric modulation, which entails forming a nanocavity beneath the gate electrode for biomolecular interaction with the dielectric material, has been extensively employed in MOSFET-based biosensors. The presence of biomolecules in these devices modifies the effective capacitance at the gate interface, resulting in a variation in channel conductance and, subsequently, the drain current. However, the efficacy of these devices is constrained by the elevated subthreshold swing and various electrostatic challenges. The vertical TFET architecture (VTFET) offers a compelling option to address these restrictions. VTFETs offer improved electrostatic control, diminished short-channel effects, and a more pronounced subthreshold swing, facilitating greater performance in biosensing applications. This study introduces a sophisticated label-free biosensing technique utilizing an InP/AlGaAs charge plasma dielectric-modulated vertical TFET (InP/AlGaAs VTFET) featuring TaN as the metal gate. This biosensor design features a nanocavity adjacent to the tunnelling junction, facilitating interactions between biomolecules and the dielectric material, resulting in fluctuations in the drain current. The device employs a twin metal gate architecture featuring laterally split dielectrics, which alleviates short-channel effects and enhances the biosensor's overall performance. The implementation of InP/AlGaAs heterostructures enhances carrier mobility, whereas the TaN gate material guarantees a high work function and stability [5], [6], [7]. This research intends to investigate the capabilities of the InP/AlGaAs VTFET biosensor for label-free biomolecule detection, emphasizing the influence of dielectric constant and charge density on the device's performance. The research examines critical metrics including energy band diagrams, drain current, drain sensitivity, and subthreshold swing (SS). Our comprehensive simulations and analyses underscore the enhanced efficacy of the proposed device, illustrating its capability for scalable, high-sensitivity biosensing applications in the identification of various biomolecules. The study specifically highlights the impact of charge density, dielectric constant, tunnel gate work function, and auxiliary gate work function on transfer characteristics and energy band diagrams, offering further insights into subthreshold swing, RF and linearity behaviour of the device. We illustrate how these aspects enhance the biosensing performance of the device through modelling and simulations. This biosensor, characterized by its heightened sensitivity, minimal power consumption, and compatibility with CMOS technology, possesses considerable potential for many biosensing applications. The suggested device design and its simulation outcomes illustrate the practicality of employing new materials and architectures to enhance the performance of TFET-based biosensors, rendering them a viable choice for future biological and environmental sensing technologies.

II. MODELLING AND SIMULATION

The proposed device architecture and fabrication methods are depicted in figures 1 and 2 for label-free biomolecule detection, utilizing a Vertical Tunnel Field-Effect Transistor (VTFET) design based on a charge plasma dielectric-modulated system. The device's core comprises InP/AlGaAs heterostructures with an intrinsic MoS_2 channel. The device incorporates a dual-gate architecture, consisting of a TaN metal gate situated above and a SiC gate positioned below, enveloped by a TiO_2 dielectric layer. The n+ - InP and p+ - InP serve as the source and drain regions, respectively, while the AlGaAs pocket region is essential for modulating the band-to-band tunnelling (BTBT) efficiency. The interface between the AlGaAs pocket region and the InP source and drain is adjusted to enhance tunnelling characteristics, enhance the device's on-state current, and minimize off-state leakage. The MoS_2 channel is selected for its exceptional electron mobility and direct bandgap, facilitating efficient charge transfer for the detection of sensitive biomolecules [8].

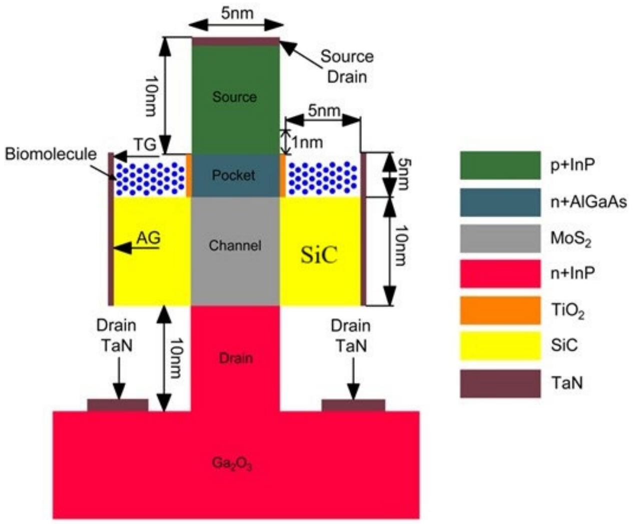

FIGURE 1. Proposed structure of dielectric modulated charge plasma based InP/AlGaAs VTFET.

The modelling parameters for the InP/AlGaAs-VTFET biosensor are presented in Table 1. The Ga_2O_3 substrate is selected for its extensive bandgap and superior thermal stability, rendering it suitable for high-performance applications across diverse operational circumstances. A TiO_2 dielectric is utilized for the gate dielectric, due to its high permittivity and ability to facilitate precise electrostatic control, which is essential for tuning the tunnelling current in the vertical direction. The TaN metal gate is used to achieve high work function alignment with the source and drain regions, ensuring efficient charge injection into the channel. In this design, a nanogap cavity is formed beneath the tunnelling gate, which traps the target biomolecules. The bio-sensing mechanism relies on changes in the dielectric constant caused by the accumulation of biomolecules in the cavity, which in turn modulates the tunnelling current. This results in band bending and threshold voltage shifts, allowing for label-free detection of biomolecules.

The device simulations are performed using Silvaco TCAD to model the electrical characteristics and bio-sensing performance. The simulations concentrate on assessing the effectiveness of band-to-band tunnelling (BTBT) and the influence of the device's gate dielectric and material composition on biosensing capability. The dual gate dielectric configuration is modelled with a high-k dielectric (TiO_2) positioned under the auxiliary gate and a SiC layer featuring an equivalent oxide thickness (EOT) beneath the tunnelling gate. The design enhances electrostatic control of the channel while diminishing gate capacitance, hence enabling efficient charge modulation for enhanced biosensing. Simulations include the I-V characteristics of the device under various biasing conditions, as well as the tunnelling current response in the presence of biomolecules. The device's performance is evaluated by essential parameters such as Energy Band Diagrams, Drain Current, Drain Sensitivity, and subthreshold swing (SS). The simulation results are analyzed to optimize the material composition, gate design, and cavity structure for achieving ultrasensitive detection limits of target biomolecules. In addition, Monte Carlo simulations are used to model the energy band fluctuations at the interface, ensuring that the device operates at optimal performance even under realistic environmental conditions, such as temperature and pressure variations [12].

III. RESULT AND DISCUSSION

When biomolecules are introduced into the cavity, the gate capacitance influences the interactions between the channel and the gate electrodes through electrostatic interactions. This alters key electrical characteristics such as the I_{ON}/I_{OFF} ratio, subthreshold swing (SS), surface potential, transfer characteristics, and energy band bending. While our previous work extensively discussed the surface potential and I_{ON}/I_{OFF} ratio, this study focuses on the subthreshold swing (SS), transfer characteristics, and energy band bending in relation to charge density (k). In this section, we observe the minimum tunnelling phenomena for an air-filled cavity (k = 1), which serves as a constant reference value throughout the literature.

A. BIOMOLECULE BINDING MECHANISM AND ADSORPTION EQUILIBRIUM

The adsorption of biomolecules onto the nanocavity surface of the InP/AlGaAs VTFET biosensor is governed by electrostatic interactions, hydrogen bonding, and van der Waals forces. When a biomolecule is introduced into the sensing region, it binds to the functionalized dielectric surface, leading to changes in the local dielectric constant and modulating the drain current. The adsorption process follows the Langmuir adsorption model, where the surface coverage (θ) depends on the biomolecule concentration (C) in solution and the adsorption equilibrium constant (K): $\theta = \frac{KC}{1+KC}$. At low concentrations, adsorption is linear with concentration, while at higher concentrations, the binding sites become saturated, reaching equilibrium. This equilibrium state ensures that the biomolecule-surface interaction is stable and repeatable, making the biosensor suitable for quantitative detection. By optimizing surface functionalization and nanocavity design, the device achieves high sensitivity and selectivity, effectively detecting biomolecules based on charge modulation and dielectric constant variations.

B. TRANSFER CHARACTERISTICS AND INFLUENCE OF CHARGE DENSITY AND DIELECTRIC CONSTANT

In this work, we investigate the feasibility of the InP/AlGaAs-VTFET biosensor with and without overlap for biosensing applications. It is well established that the entry of biomolecules into nanogaps induces a dielectric modulation effect. In our proposed InP/AlGaAs-VTFET biosensor, the nanogaps under the tunnel gate are filled with insulating material that has an equivalent dielectric constant corresponding to the biomolecule under study, as represented

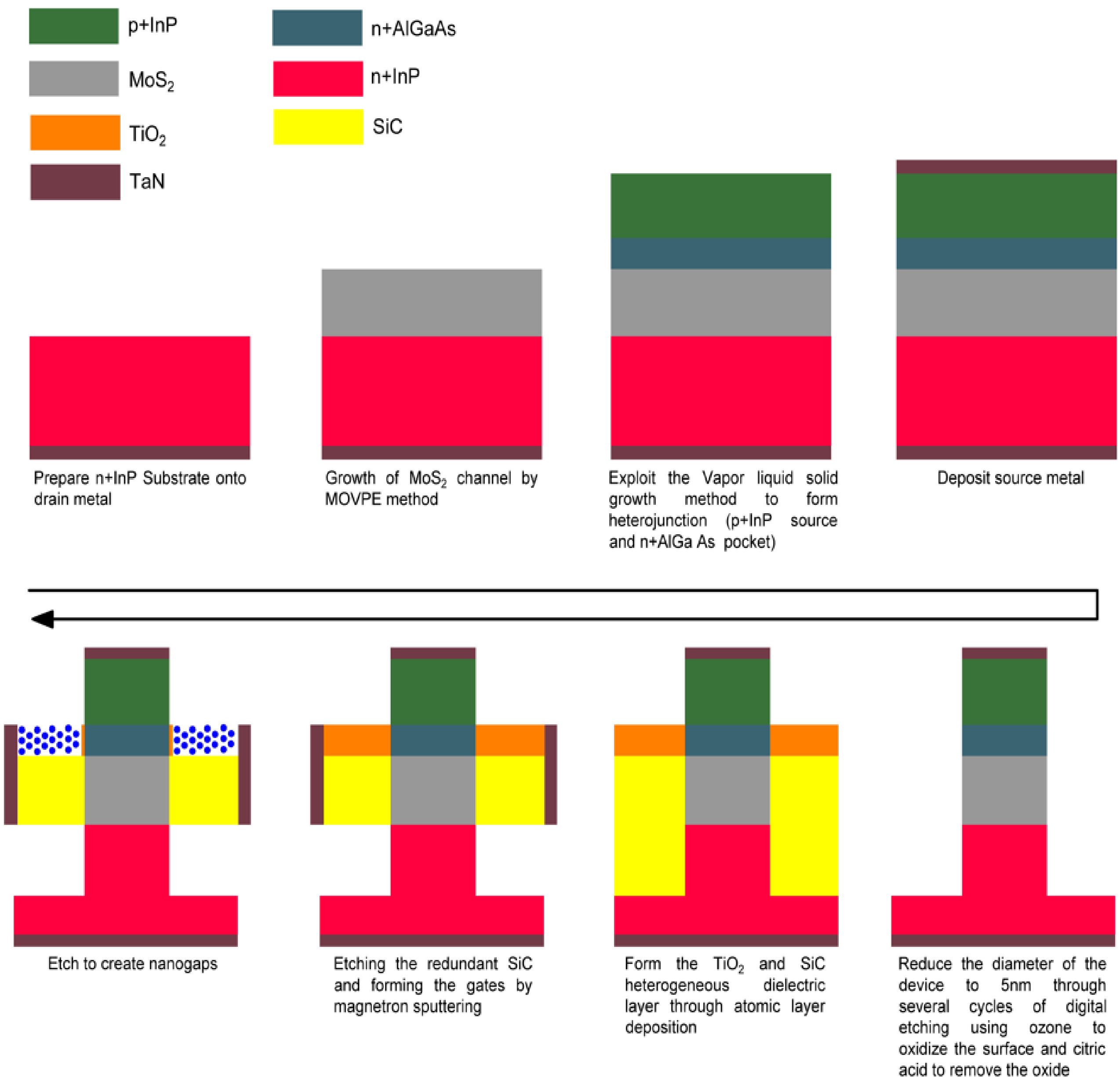

FIGURE 2. Proposed fabrication flow diagram.

in our simulation. To account for the impact of charged biomolecules, we define interface states between the high-k dielectric material (TiO_2) and the equivalent biomolecule material. To enhance simulation accuracy, Figure 3(a) presents the model calibration of the InP/AlGaAs-VTFET biosensor, where the simulation results align well with the experimental data from reference [10], demonstrating good consistency between the modelled and experimental transfer characteristics. Figure 3(b) shows the InP/AlGaAs-VTFET biosensor's transfer characteristics with and without overlap, where Vds is 0.45V and Vgs is a voltage between the gate and the source that varies between 0V and 1.5V in increments of 0.025V. Air (k = 1) fills the nanogaps in these models. The results demonstrate that the devices' OFF-state current (Vgs = 0V) and ON-state current (Vgs = 1V) are highly comparable. To be more precise, the switching ratio is around 109 because the OFF-state current for one device is 10^{-17} A and the ON-state current for the other is 10^{-9} A. Moreover, the average subthreshold swing for both devices is found to be 60 mV/Dec, which is an excellent value for the biosensing application. Although the ON-state current of the InP/AlGaAs-VTFET biosensor without overlap is slightly higher than that of the device with overlap, the current sensitivity of the device with overlap is significantly better. This enhanced sensitivity will be discussed in greater detail in the following sections [13], [14], [15].

TABLE 1. **Simulation parameters of InP/AlGaAs-VTFET biosensor and dielectric constant of different biomolecules [9], [10], [11].**

Parameters in nm	Parameter Value in nm
Source Length (s)	5
Substrate Length (Lsub)	20
Source Height (Hs)	10
Pocket Height (Hp)	2.5
Channel Height (Hc)	10
Drain Height (Hd)	10
Tunnel Gate Height (Ht)	10
Auxiliary gate Height (Hag)	10
TiO2 Thickness	5
Sic Thickness	5
TiO2 Length	10
Sic Length	10
Work function of Tunnel and Auxiliary gate	4.2 and 4.0 eV
Biomolecules	Dielectric Constant (K)
(3-Aminopropyl) triethoxysilane (APTES)	4
Keratin	8
Gelatin	12
Albumin	7

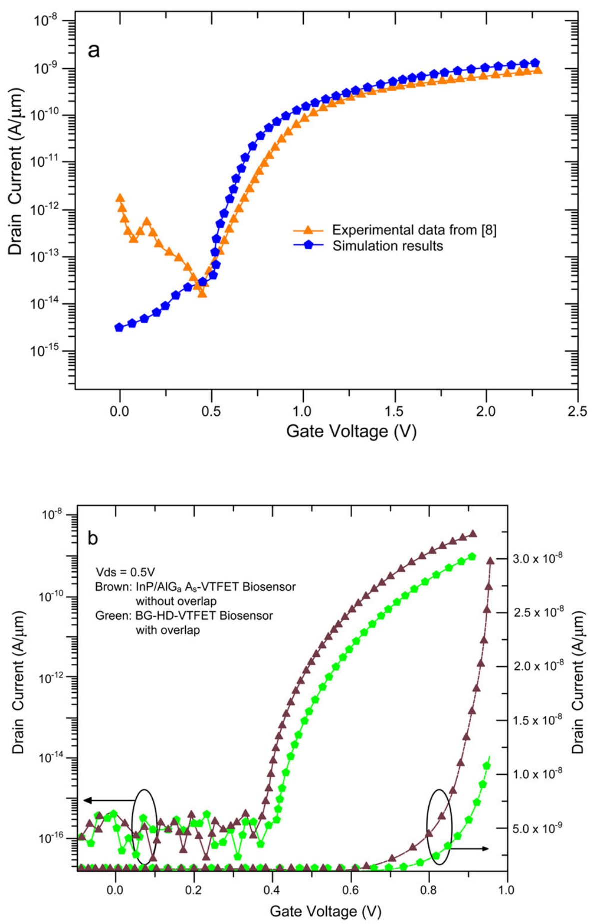

FIGURE 3. **(A) Simulation models calibrated to replicate the findings of reference [10]; (b) Transfer characteristics of the InP/AlGaAs-VTFET biosensor with and without overlap.**

Several key parameters are used to evaluate the effectiveness of biosensors, such as drain current, threshold voltage, and subthreshold swing, with drain current being the most commonly utilized parameter. As reported in various references [9], [10], [11], the current sensitivity of the biosensor is calculated using the following formula:

$$I_s = \frac{I_{d,k}}{I_{d,air}} \tag{1}$$

where $I_{d,k}$ is the drain current of the biosensor with biomolecules and $I_{d,air}$ is the drain current of the biosensor without biomolecules. Another important parameter for assessing biosensor performance is threshold voltage sensitivity, which is defined as:

$$v_s = \frac{v_{air}-v_k}{v_{air}} \tag{2}$$

where v_k is the threshold voltage of the biosensor with biomolecules, and v_{air} is the threshold voltage of the biosensor without biomolecules at which the drain current reaches (1.0×10^{-8}A/μm). At a constant dielectric constant (k = 7), Figure 4 (a) and Figure 4 (b) display the transfer characteristics of the InP/AlGaAs-VTFET biosensor with and without overlap, for positive and negative charge densities. The performance of the biosensor shows comparable trends of variation for various charge densities for devices with and without overlap. However, the degree of change is more significant in the InP/AlGaAs-VTFET biosensor with overlap compared to the device without overlap.

For instance, the ON-state current of the biosensor with overlap increases from 9.5 to 9.8 A/μm as the charge density increases from -5.0 $\times$ 10^{12} cm^2 to 5.0 $\times$ 10^{12} cm^2, resulting in a rise by a factor of 10^{45}. In contrast, the ON-state current of the InP/AlGaAs-VTFET biosensor without overlap increases from 2.5 $\times$ 10^{-8} A/μm to 1.5 $\times$ 10^{-8} A/μm for the same charge density range, which corresponds to a rise by a factor of 535. The current sensitivity of the overlapped device is much superior, even though the ON-state current of the non-overlapping device is larger for some charge densities. The trends indicated in Figures 4 (a) and (b) are congruent with the fluctuation in ON-state current and current sensitivity as a function of charge density, as shown in Figure 4. (c). Without overlap, the InP/AlGaAs-VTFET biosensor has a maximum current sensitivity of just (1.8 $\times$ 10^3), however with overlap, the device has a substantially greater maximum current sensitivity of (3.5 $\times$ 10^3). The threshold voltage sensitivity of the InP/AlGaAs-VTFET biosensor with overlaps grows from 0.36 to 0.77 as the charge density increases from -5.0×10^{12} cm^2 to 5.0 $\times$ 10^{12} cm^2, while the threshold voltage sensitivity of the device without overlaps increases from 0.28 to 0.78. The potential distribution for the charge densities (rho = 5 $\times$ 10^{12}) in the InP/AlGaAs-VTFET biosensor with overlap is noticeably bigger than in the device without overlap when the charge densities are the same, as illustrated in Figure 4. (d). The behaviour shown in Figures 4 (a) and 4 (b) is explained by the fact that the potential increases as the charge density does. According to Figure 4. (d), the potential curves of

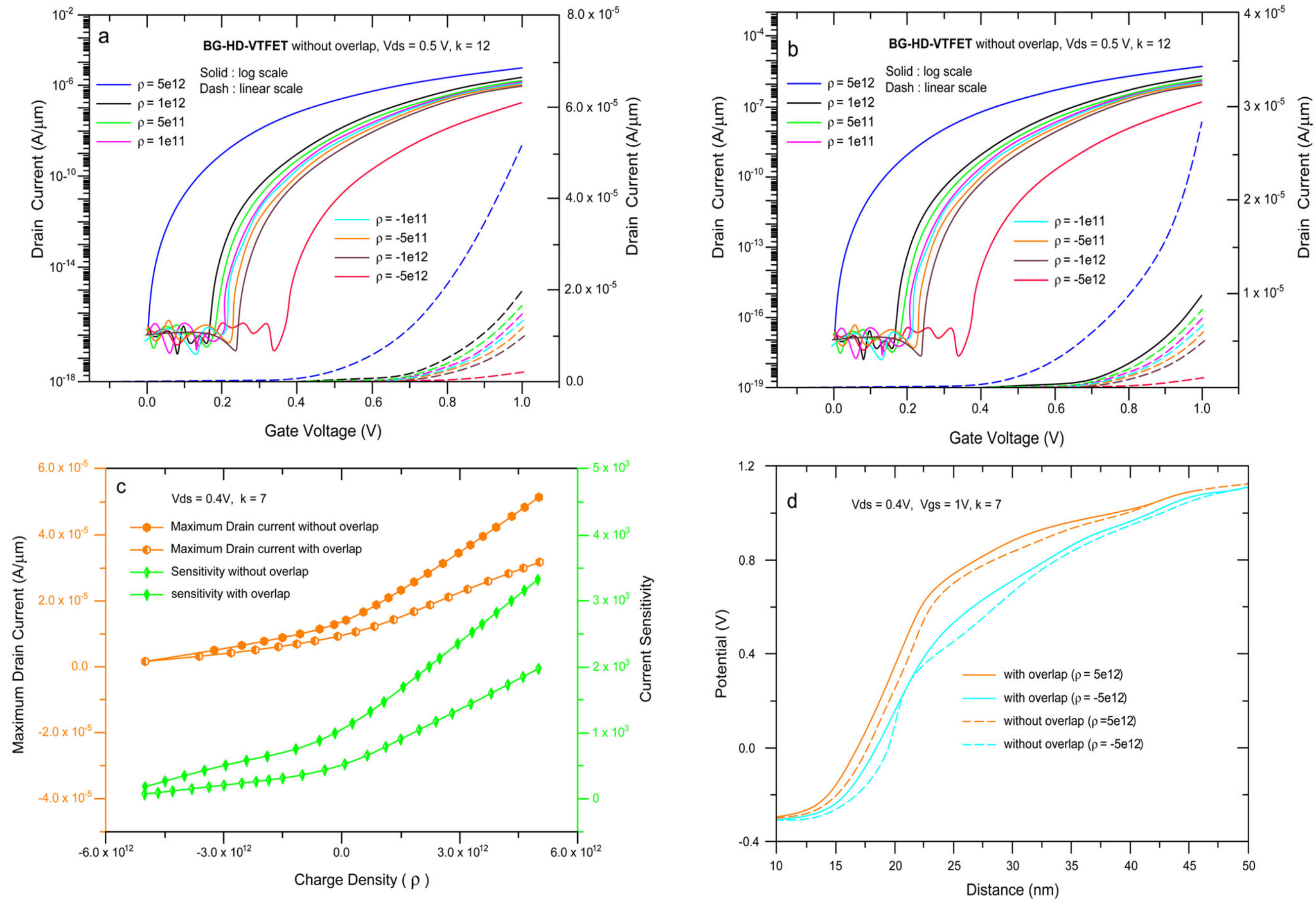

FIGURE 4. (a) Transfer characteristics of InP/AlGaAs-VTFET without overlap for different charge density; (b) Transfer characteristics of InP/AlGaAs-VTFET with overlap for different charge density; (c) the variation of sensitivity and maximum ON-state current for different charge density; (d) the potential distribution of InP/AlGaAs-VTFET without and with overlap.

the devices with and without overlap are very close to each other. Figure 4. (e) presents the potential and electric field distribution for the InP/AlGaAs-VTFET biosensor without overlap at different charge densities. The electric field distribution follows the potential distribution, and as charge density increases, so does the potential. Because the electric field is stronger close to the pocket area because of the increased charge density, more electrons are able to tunnel from the source to the pocket.

C. IMPACT OF TUNNEL GATE WORK FUNCTION AND AUXILIARY GATE WORK FUNCTION

Following this, we investigate how the InP/AlGaAs-VTFET biosensor's detection performance is affected by the tunnel gate and auxiliary gate work functions.

The InP/AlGaAs-VTFET biosensor's transfer characteristics, both with and without overlap, are shown in Figure 5(a) for different values. Increasing from 3.7 eV to 4.7 eV in increments of 0.10 eV, the OFF-state current of both devices stays at a low value, as demonstrated. But when the voltage increases, the current through the ON-state of both devices drops. Under various conditions, the InP/AlGaAs-VTFET's ON-state energy band diagrams without overlap (Figure 5(b)) and with overlap (Figure 5(c)) are displayed. As the gate function increases, the conduction band grows closer to the tunnel gate, while the valence band grows at a slower rate. The ON-state current drops as a result of a smaller effective tunnelling area at the source and pocket interface, as shown in Figure 5(c). Because of the stronger electric field close to the tunnel gate, the current change in the overlapped InP/AlGaAs-VTFET is also more noticeable. While the InP/AlGaAs-VTFET with overlap has a tunnel length of 5.0 nm at a metal gate function of 4.0 eV, the InP/AlGaAs-VTFET without overlap has a tunnel length of 5.50 nm. Furthermore, the threshold voltage of both devices increases with an increase in gate function. Figure 5(d) shows the current sensitivity of the InP/AlGaAs-VTFET biosensor without overlap and with overlap under different values of gate function. It can be observed that the current sensitivity of both devices decreases with an increase in gate function, however, the current sensitivity of the InP/AlGaAs-VTFET with overlap remains significantly higher than that of the InP/AlGaAs-VTFET without overlap.

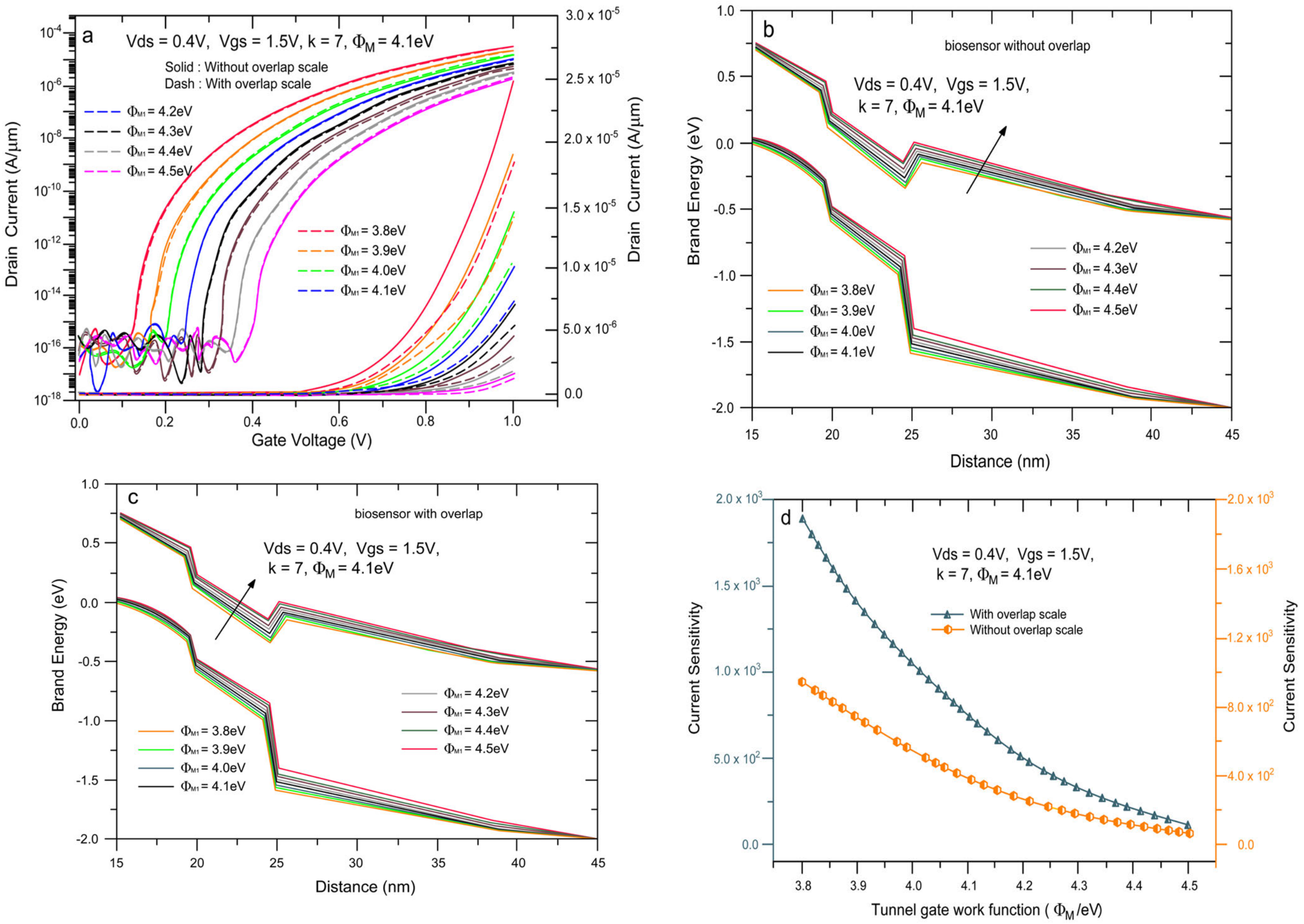

FIGURE 5. **(a) Transfer characteristics of InP/AlGaAs-VTFET with and without overlap; (b) the variation of the ON-state energy band diagram of InP/AlGaAs-VTFET without overlap; (c) the variation of the ON-state energy band diagram of InP/AlGaAs-VTFET with overlap; (d) sensitivity and ON-state current of InP/AlGaAs-VTFET under varying tunnel gate work functions.**

Additionally, the threshold voltage sensitivity of both devices decreases as gate function increases.

D. EFFECT OF DIELECTRIC CONSTANT ON ENERGY BAND DIAGRAMS, DRAIN CURRENT, DRAIN SENSITIVITY AND SUBTHRESHOLD SWING (SS)

A significant potential barrier is evident at the source-channel junction of the InP/AlGaAs-VTFET biosensor when in the off state (with VDS = 0.50 V and VGS = 0 V), as depicted in Figure 6(a). The conduction band of the channel is misaligned with the valence band of the source. The energy bands align, resulting in the formation of a minor tunnelling barrier when the device transitions to the ON state (VDS = 0.50 V and VGS = +1.50 V). Consequently, electrons can traverse the conduction band of the channel and enter the valence band of the source. Outcomes are generated using a continuous cavity length of 10 nm and a positively charged biomolecule with a dielectric constant (k = 12). Figure 6(b) illustrates the band diagram for the ON and OFF states, corresponding to both positively and negatively charged biomolecules, with a constant relative permittivity (k = 7.0) for a cavity length of 10 nm. The graph shows that positive charged biomolecules cause the energy band to shift downwards, while negative charged biomolecules shift the energy band upwards. For positive charged biomolecules, the gate capacitance exerts stronger electrostatic control over the intrinsic channel, promoting earlier tunnelling effects and enhancing the tunnelling phenomenon. In contrast, a fully filled cavity with negatively charged biomolecules results in a more gradual misalignment of the energy bands, which reduces the tunnelling effects in the ON state, leading to a drop in the drain current [16], [17], [18].

The drain characteristics of the InP/AlGaAs-VTFET biosensor for neutral biomolecules with a fixed relative permittivity (k = 6.5) and varying cavity lengths are shown in Figure 7(a). For cavity lengths greater than 5 nm, the length does not significantly affect the drain characteristics, as the effective metal work function over the junction area between the channel and source is sufficient to trigger lateral tunnelling. For shorter cavity lengths (5 nm or less), the channel offers more resistance to tunnelling until it reaches

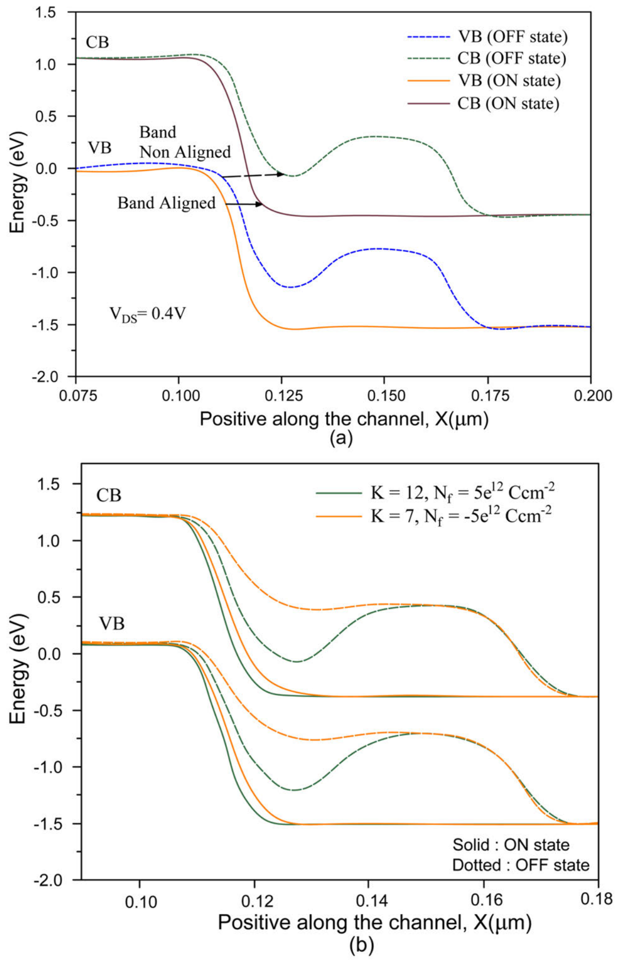

FIGURE 6. **Band diagram for energy during the ON and OFF state conditions (a) Effect of the neutral biomolecule dielectric constant k = 12. (b) Comparison of energy bands for biomolecules with negative and positive charges for the same dielectric constant k = 7.**

the threshold voltage (1.0 V), due to a smaller coverage area of the lower work function metal. To enhance tunnelling, a larger area of the lower metal work function gate is needed. The OFF current (IOFF) remains unchanged as ambipolarity behaviour is reduced due to the dual metal gate structure. For larger cavity lengths, such as (Lc = 15) nm, ambipolarity decreases, reducing band bending and effectively lowering IOFF. The optimal cavity length for further investigation is chosen as Lc = 10nm. Figure 7(b) shows drain characteristics for different biomolecules. As the dielectric constant of the biomolecule increases, the energy barrier between the source and channel narrows, increasing the ON current (I_{ON}). Biomolecules thus modulate the tunnelling process, and higher dielectric constants result in stronger electrostatic coupling, enhancing I_{ON}. For positively charged biomolecules in the cavity, with (VDS=0.5V and (k = 7), Figure 7(c) shows an increase in drain current as the positive charge strengthens the tunnelling effect. IOFF remains unchanged as the channel-drain junction is unaffected. Conversely, for negatively charged biomolecules, Figure 7(d) illustrates a decrease in I_{ON} as the negative charge widens the tunnelling barrier [19], [20], [21], [22], [23].

Drain sensitivity is a key parameter for detecting various biomolecules in the InP/AlGaAs-VTFET biosensor. It is defined as the ratio of the drain current when the cavity is fully occupied by biomolecules to the drain current, when the cavity is vacant or filled with air:

$$d_s = \frac{i_{d(bio)}}{i_{d(air)}} \tag{3}$$

The drain sensitivity is analyzed for fixed Vds = 0.50V, with varying gate voltage, charge, cavity length, and dielectric properties. The sensitivity is higher for positively charged biomolecules, as shown in Figure 8(a). Increased positive Researchers have discovered that a cavity length of about 10 nm produces the best drain sensitivity. An additional benefit of increasing the biomolecules' dielectric constant (k) is an improvement in drain sensitivity. As seen in Figure 8(d), the limit of drain current sensitivity is reached at 3.0×1^{09} with a dielectric constant of (k = 7). A larger drain current and increased sensitivity result from an increased likelihood of tunnelling between the source and the channel [24].

To identify specific biomolecules, we use the "Selectivity" parameter (ΔS), which is defined as:

$$\Delta_s = \frac{i_{d(k2)} - i_{d(k1)}}{i_{d(k1)}} \tag{4}$$

where (k2) and (k1) are the dielectric constants of two different biomolecules. For instance, the selectivity between albumin (with (k = 7)) and bacteriophage T7 (with (k = 6.3)) is:

$$\Delta_s = \frac{i_{d(7)} - i_{d(6.3)}}{i_{d(6.3)}} = 1.2708 \tag{5}$$

Similarly, for the pair gelatin (k = 12) and keratin (with k = 8k) is: the selectivity is calculated as:

$$\Delta_s = \frac{i_{d(12)} - i_{d(8)}}{i_{d(8)}} = 22.1 \tag{6}$$

This selectivity ensures that the biomolecule binding process, based on dielectric modulation, allows for the accurate transduction of the analyte's presence into a proportionate modification of the output electric signals. A complete analysis of selectivity can be obtained through transient analysis of the InP/AlGaAs-VTFET biosensor. The impact of neutral biomolecules on the subthreshold swing (SS) of the InP/AlGaAs-VTFET biosensor is demonstrated in the bar plot in Figure 9, for a cavity length Lcav = 10 nm and Lcav = 20nm. When the relative permittivity of the neutral biomolecules is varied from k = 3.57 to k = 12, the SS decreases from 30 mV/dec to 20 mV/dec, while the air-filled cavity shows a higher SS of 75 mV/dec. The reduction in SS with increasing dielectric constant is attributed to the subtle effect of the dielectric material on the tunnelling junction, which enhances tunnelling current, while the dual-metal work function further allows the device to switch more rapidly

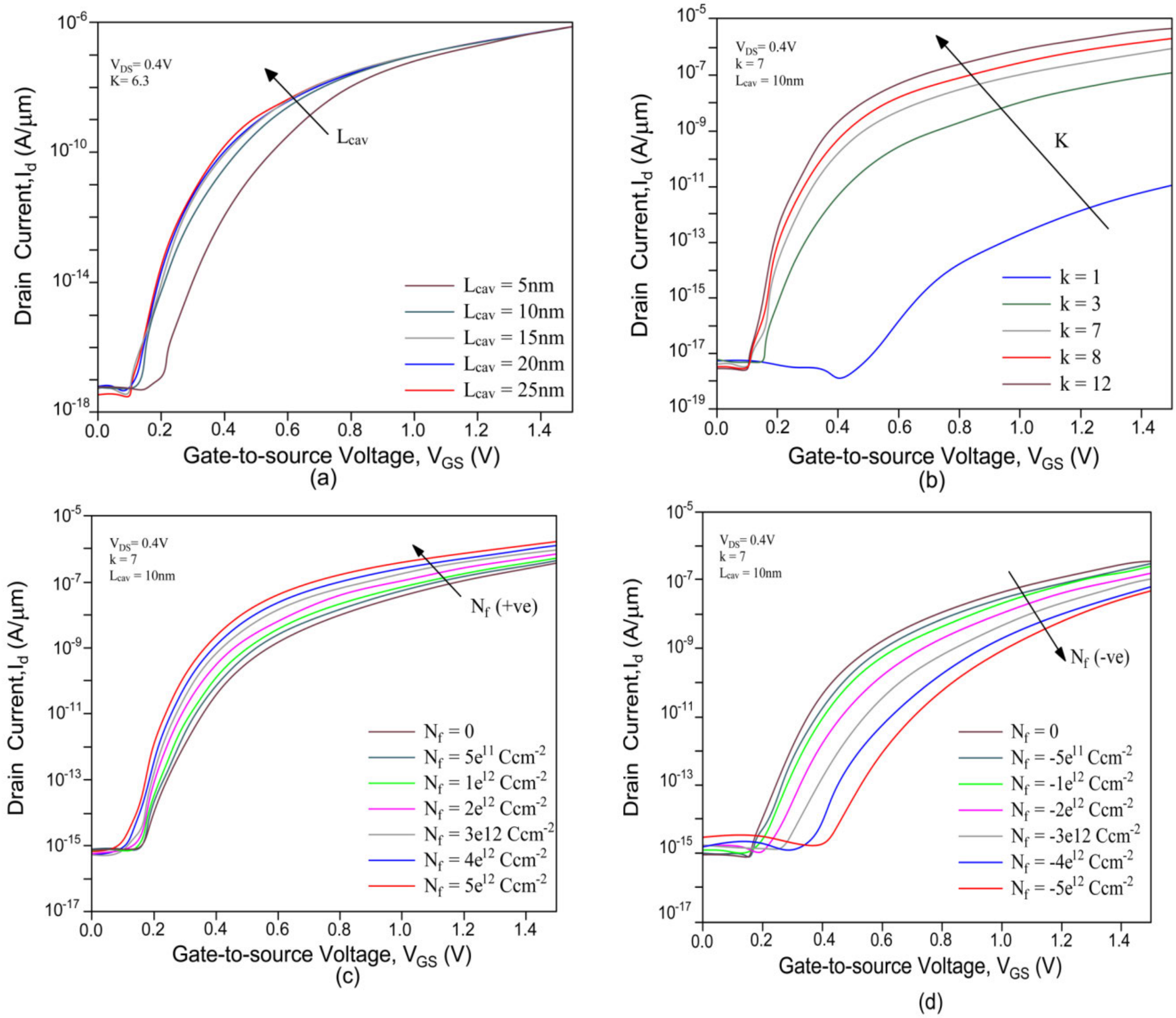

FIGURE 7. Transfer characteristics of CP DM DGTFET with a fixed VDS = 0.5 V (a) Cavity length, Lcav = 6, 12, 18, 20 and 25 nm (b) Dielectric constant (k = 1, 3.57, 6.3, 8, and 12) (c) Positive charge distribution (ranging from 0 Ccm-2 to 10-12 Ccm-2) for k = 5. (d) Negative charge distribution (ranging from 0 Ccm-2 to -10-12 Ccm-2) for k = 5.

from the ON to OFF state. The lowest SS of 20 mV/dec is achieved at k = 12, indicating improved switching performance. As shown in Figure 9, the device operates effectively at a low supply voltage VDS = 0.5. The threshold voltage of the tunnel FET, Vt and the voltage Voff are essential in determining the device's switching characteristics. The drain current Id (Voff) refers to the OFF-state current, while Id (Vt) corresponds to the current at the threshold voltage. The influence of neutral biomolecules on the average subthreshold slope (AVSS) has been analyzed, and the results are shown in Figure 10(a). It is observed in Figure 10(a) that the I_{ON}/I_{OFF} ratio increases with higher k-values of the biomolecules, further enhancing the device's switching performance. Moreover, the AVSS decreases rapidly for k-values less than 5, suggesting that smaller SS values correlate with a better electrical response and improved detection capabilities. For charged biomolecules, the I_{ON}/I_{OFF} ratio and the average subthreshold slope (AVSS) are plotted in Figure 10(b) with respect to the variation in charge density, from negative to positive. At the tunnelling junction, as the positive charge increases, the ON current is enhanced due to the steep band bending of the energy bands, which reduces the tunnelling barrier. Consequently, both the I_{ON}/I_{OFF} ratio and the AVSS increase, indicating an improvement in the sensor's performance with higher positive charge biomolecules [25], [26], [27].

E. RF AND LINEARITY ANALYSIS

To keep the InP/AlGaAs-VTFET biosensor's dynamic power consumption low, it is crucial to evaluate its RF performance for different biomolecules. In this study, we examined

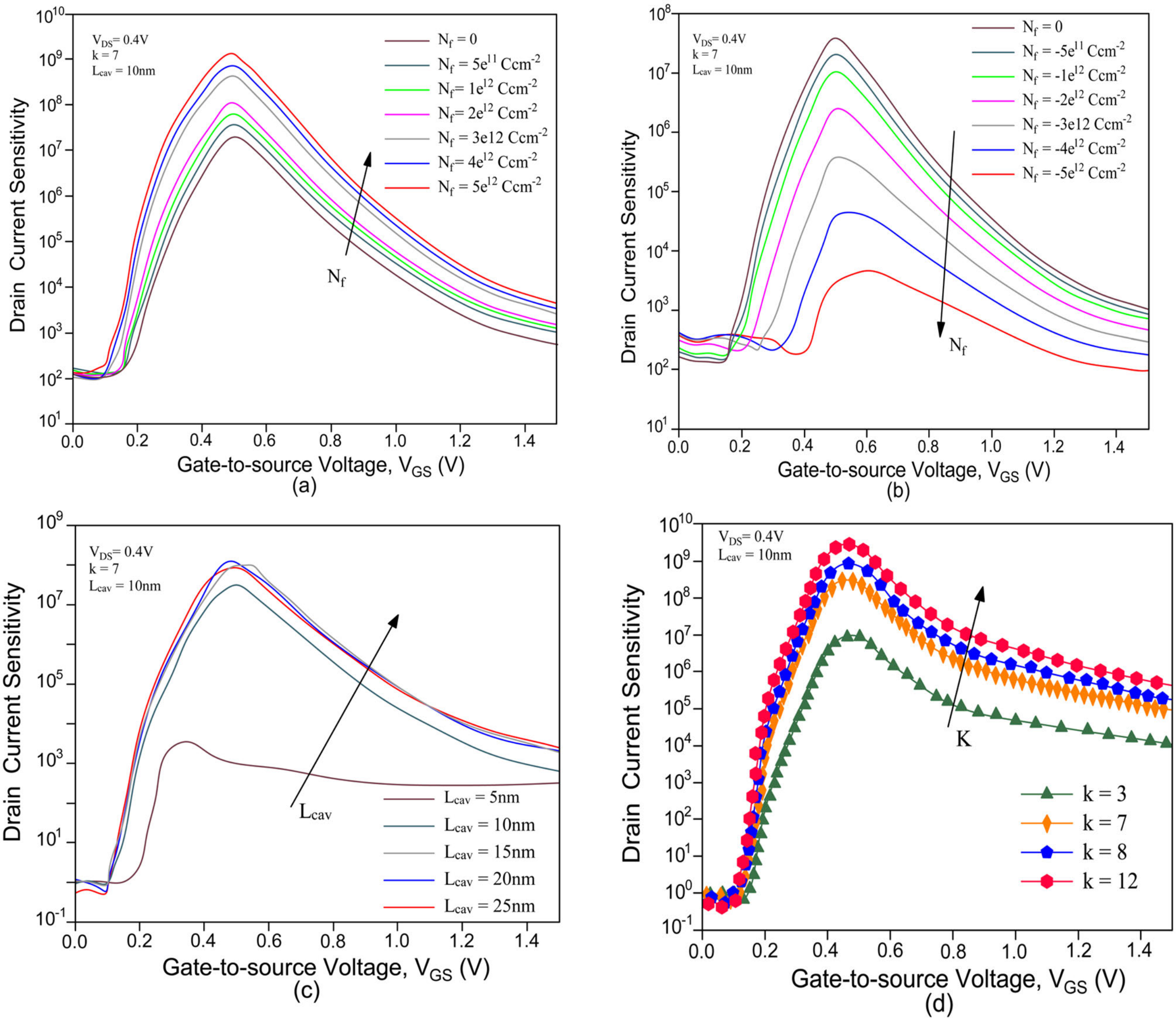

FIGURE 8. Show the plot of drain sensitivity at fixed VDS =0.5 V with variation of (a) Density of positive charges (b) Density of negative charges (c) Cavity length for neutral biomolecule with k = 6.3 (d) Relative permittivity of various neutral biomolecules (k = 3.57, 6.3, 8, 12) for Lcav = 10 nm.

important RF parameters including gain bandwidth product (GBP), transconductance (gm), gate-to-drain capacitance (Cgd), and cutoff frequency (fT). Biosensing sensitivity is greatly affected by the transconductance (gm), which is a measure of the device's ability to convert gate voltage into drain current. Parasitic capacitances like the gate-to-drain capacitance (Cgd) affect circuit power dissipation and switching behaviour, both of which are crucial in radio frequency (RF) applications. Cgd is defined as follows:

$$c_{gd} = \frac{\gamma_{qg}}{\gamma_{vds}} \tag{7}$$

where qg is the gate charge. Figure 11(a) illustrates that the transconductance with neutral biomolecules increases with higher dielectric constants; gm demonstrates a nearly linear dependence on gate voltage for all k-values when VGS>0.7 V. The variation of Cgd with gate voltage for different biomolecules is plotted in Figure 11(b), indicating that an increase in dielectric constant results in reduced capacitive coupling at the channel-drain region. This effect is due to the lower density of states, which is advantageous for high-frequency operation [27]. As gate voltage increases, charge builds up at the drain junction, leading to an increase in total parasitic capacitance. At around 0.7 V, there is a rapid rise in gm, highlighting the device's sensitivity at this threshold. The cutoff frequency (fT), which denotes the maximum frequency at which the device can act as a unity-gain current amplifier, is another important parameter for high-speed switching applications. It is given by:

$$f_t = \frac{g_m}{2\pi c_{gg}} \tag{8}$$

where $c_{gs} = c_{gg} + c_{gd}$ and c_{gs} is the source-gate capacitance. Figure 11(c) shows the variation of fT with neutral

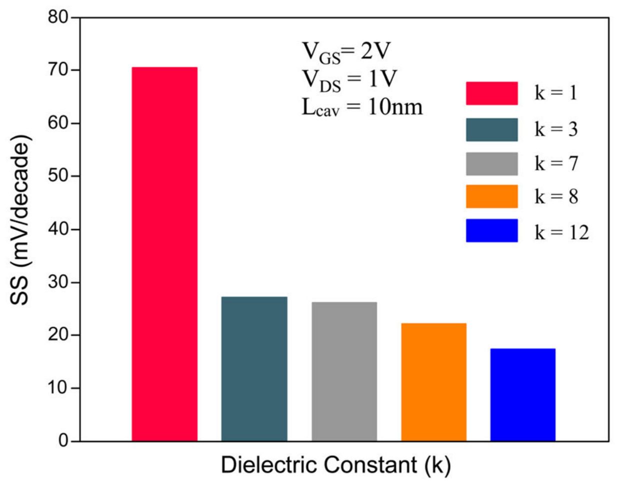

FIGURE 9. Subthreshold slope (SS) of various neutral biomolecules for VGS = 1.5 V, VDS = 0.5 V and Lcav = 20 nm.

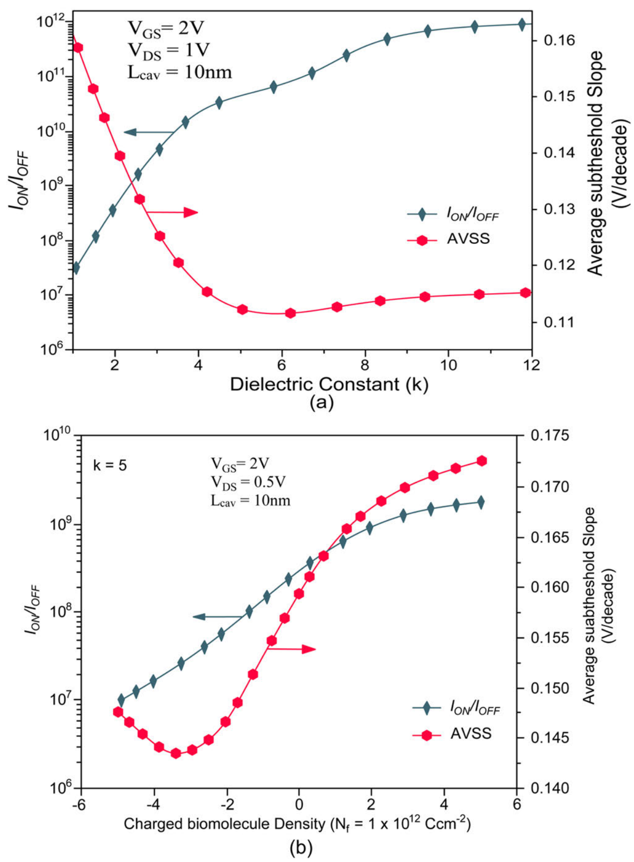

FIGURE 10. Variation of AVSS, and ION/IOFF ratio of Charge Plasma based DM DGTFET for (a) Neutral biomolecules for Lcav = 20 nm (b) Charged molecules of constant permittivity k = 5 and Lcav = 20 nm.

biomolecules present in the cavity. The tunnelling at the source-channel region enabled by high-k dielectric materials increases the cutoff frequency, and the near-linear rise in fT with gate voltage (>0.7 V) aligns with the trend in Cgd shown in Figure 11(b). As gm continues to increase more sharply for the InP/AlGaAs-VTFET biosensor, parasitic capacitance builds up at the channel-drain region until VGS = 1V, resulting in a further linear increase in fT. In the absence of a biomolecule (k = 1), there is a significant tunnelling gap at the source junction, leading to minimal hole tunnelling. This increase in cutoff frequency enhances biomolecule detection capability. The gain bandwidth product (GBP) exhibits behaviour similar to fT, as shown in Figure 11(d). GBP, defined as the product of gain and bandwidth, represents the device's operational region within a stable gain regime, suitable for high-frequency applications:

$$gbp = \frac{g_m}{2\pi c_{gd}} \tag{9}$$

Another measure for high-speed operation is the transit time (t_t), defined as the time required for charge carriers to traverse the channel region:

$$t_t = \frac{1}{2\pi f_t} \tag{10}$$

A shorter transit time enhances the device's switching speed and overall performance. Transit times (t_t) for different neutral biomolecules are shown in Figure 12. The InP/AlGaAs-VTFET biosensor demonstrates faster sensing speeds for high-k dielectric biomolecules such as albumin. With charge carriers moving through the channel, the high dielectric constant narrows the tunnelling barrier as gate voltage increases, further improving the device's performance in biomolecule detection.

Minimizing signal distortion is essential for preserving signal integrity and quality in high-frequency carrier transport. Therefore, a linearity analysis was conducted to evaluate the performance of the InP/AlGaAs-VTFET biosensor in the RF regime. The linearity of the device is examined using parameters such as the voltage intercept point (VIP), input intercept point (IIP), and intermodulation distortion (IMD), defined as follows:

$$g_{mn} = \frac{1}{\mathrm{n!}}\frac{I_{\mathrm{d}}^{\mathrm{n}}}{\delta \mathrm{V_{GS}}} \tag{11}$$

$$V_{\mathrm{IP2}} = 4\left(\frac{g_{m1}}{g_{m2}}\right) \tag{12}$$

$$V_{\mathrm{IP3}} = \sqrt{24 \times \left(\frac{\mathrm{g_{m1}}}{\mathrm{g_{m3}}}\right)} \tag{13}$$

$$\mathrm{IIP3} = \frac{2}{3} \times \left(\frac{\mathrm{g_{m1}}}{\mathrm{g_{m3}} \times \mathrm{R_s}}\right) \tag{14}$$

$$\mathrm{IMD3} = \left[\frac{9}{2} \times (\mathrm{VIP3})^2 \times \mathrm{g_{m3}}\right]^2 \times \mathrm{R_s} \tag{15}$$

where (gm) represents transconductance, (n) is the order (where (n = 1, 2, 3) and (Rs) is typically set to 50 Ω for RF applications. Higher-order parameters such as VIP2, VIP3, and IIP3 should be maximized to achieve improved linearity across different DC operating conditions. Figures 13(a), (b), and (c) depict the variations

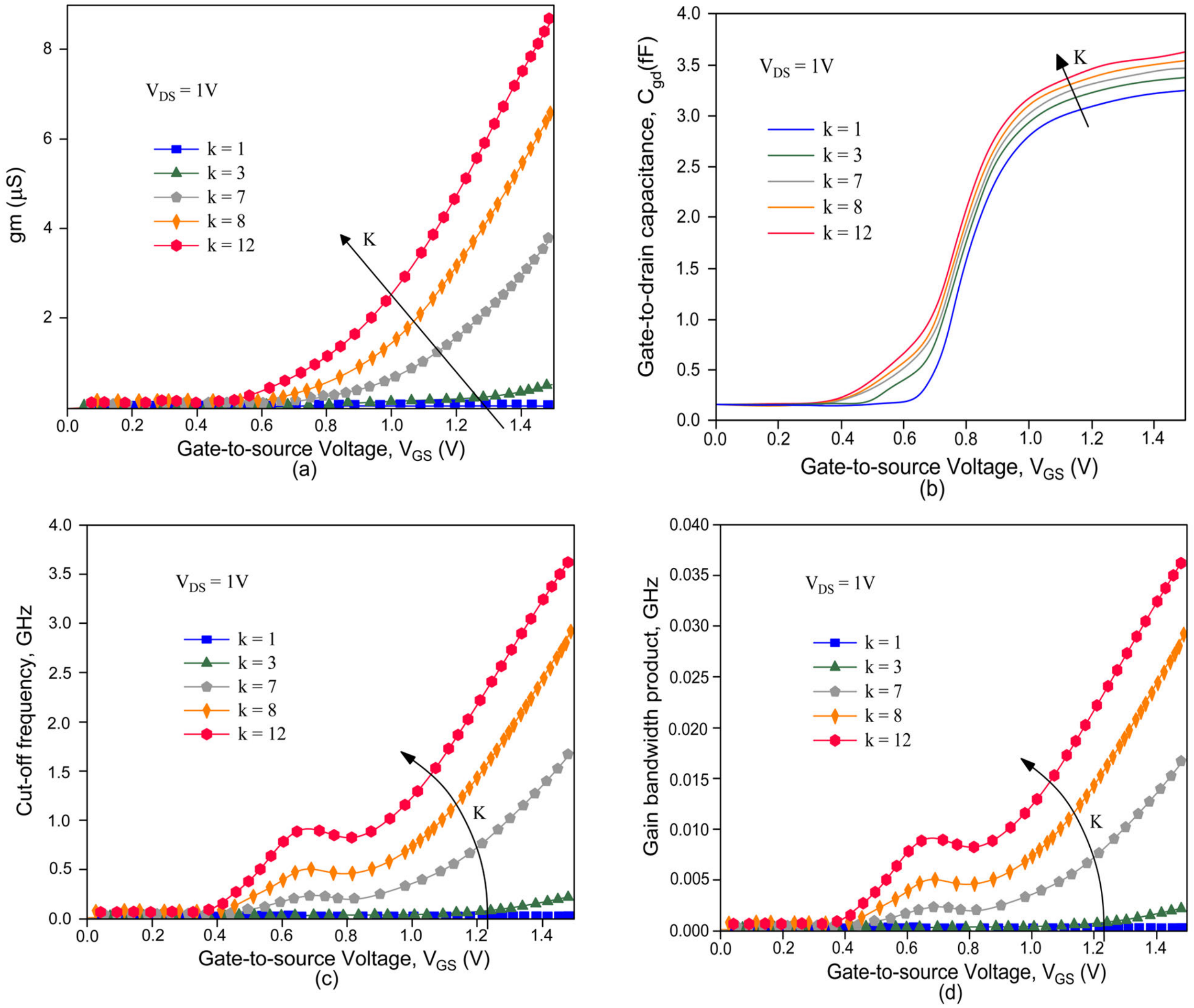

FIGURE 11. RF analysis with constant VDS = 0.5 V (a) gm (b) Cgd (c) fT (d) GBP for various neutral biomolecule.

of VIP2, VIP3, and IIP3 with gate-to-source voltage for various biomolecules, with a fixed drain-source voltage of (VDS = 0.5V).

Biomolecules with a higher dielectric constant (k) display enhanced linearity metrics, as shown in Figure 13(a). Higher (k) biomolecules enable greater tunnelling, leading to a stabilized (gm) value, thus yielding a more linear response. VIP3 is particularly significant as it includes the third-order transconductance (gm). Figure 13(b) illustrates VIP3 values for biomolecules with (k) values of 1, 3.57, 6.3, 8, and 12. In the air-filled cavity (where (k = 1), VIP3 is shown in subplot (i) of Figure 13(b) and so forth. The variation of VIP3 with gate bias reveals distinct peaks at different biases for each biomolecule. When the cavity is filled with a high (k) biomolecule, the device demonstrates reduced distortion [28], [29]. This nonlinearity is further assessed using IIP3, as shown in Figure 13(c), which exhibits peaks beyond the saturation point for all tested biomolecules. The variation of IMD3 with input gate voltage, displayed in Figure 13(d), indicates that the device maintains a robust performance with minimal distortion. The drain current values obtained with the InP/AlGaAs-VTFET biosensor for different biomolecules were compared to previous biosensor studies, as shown in Table 3. The performance of this biosensor for neutral biomolecules within the cavity surpasses that of other biosensors in sensitivity at lower operating voltages, underscoring that this InP/AlGaAs-VTFET device achieves higher sensitivity and improved performance for biomolecule detection.

F. DEVICE PHYSICS AND ELECTROSTATIC ANALYSIS

The underlying physics of the proposed InP/AlGaAs charge plasma dielectric-modulated vertical TFET (VTFET) is governed by charge plasma modulation, band-to-band tunneling, and biomolecule-induced dielectric variations. To enhance

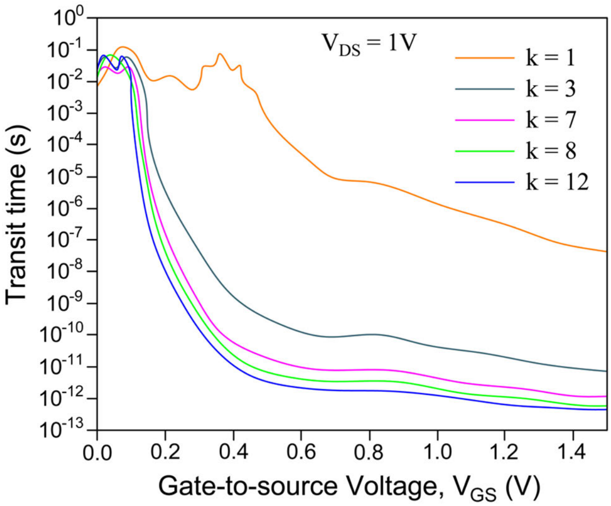

FIGURE 12. Variation of tt with VGS for various neutral biomolecules at constant VDS = 0.5 V.

TABLE 2. Comparison of InP/AlGaAs-VTFET-based biosensor performance with previously reported works.

K Value of Target Biomolecules	Drain Current, I_{ON} & I_{OFF} ratio	Present Work	Other works reported	References
Neutral 3.6	Vgs = 1.5V Vds = 1V	1.2×10^{-5}	5.1×10^{-10}	[11]
Neutral 6.3	Vgs = 1.5V Vds = 1V	5.2×10^{-5}	2.1×10^{-8}	[12]
Neutral 8	Vgs = 1.5V Vds = 1V	1.2×10^{-5}	9.1×10^{-8}	[13]
Neutral 12	Vgs = 1.5V Vds = 1V	2.3×10^{-5}	5.5×10^{-7}	[14]
Neutral 12	Vgs = 1.5V Vds = 1V	9×10^{10}	4×10^{-7}	[15]
Neutral 12	Subthreshold Swing (SS)	16mV/dec	2.17mV/dec	[16-18]
K=6.3	Vgs = 1.2V Vds = 1V	4.9×10^{-5}	3.8×10^{9}	[19]
K-12	Vgs = 1.2V Vds = 1V	2×10^{-5}	3.7×10^{8}	[20-21]
K-8	Vgs = 1.2V Vds = 0.5V	3.5×10^{-5}	3.9×10^{-7}	[22-23]
K=7	Vgs = 1.2V Vds = 0.5V	8×10^{-5}	9.1×10^{-7}	[24-25]

tunneling efficiency, the device integrates a nanocavity near the tunneling junction, allowing biomolecule interactions to induce fluctuations in the local charge density and electrostatic potential. In the proposed design, the charge plasma effect is utilized to improve carrier transport and electrostatic control. The twin-metal gate structure induces a localized charge plasma region in the InP/AlGaAs heterostructure, which modulates the energy band profile at the tunneling junction. The high-k dielectric layer within the nanocavity alters the surface potential depending on the dielectric constant (k) of the biomolecule, influencing the drain current and sensitivity. Following the methodology in [9] and [10], charge plasma TFETs exhibit tunable electrostatics, which is further enhanced in our work through: Optimized nanocavity placement, ensuring strong interaction between biomolecules and the tunneling region, Source-pocket overlap, improving tunneling efficiency and reducing subthreshold swing (SS), Lateral dielectric split, mitigating short-channel effects (SCEs) and improving control over charge transport. Figure 14 illustrates the energy band diagrams under different biomolecule conditions, demonstrating how dielectric variations influence band bending and tunneling probability.

The energy band diagrams in Figure 14 show the modulation of conduction and valence band edges under different biomolecule conditions. Similar to the band engineering techniques discussed in [9] and [10], our results indicate: A steeper band bending at the source-channel junction due to enhanced charge plasma formation, Higher band-to-band tunneling (BTBT) probability, improving device sensitivity to biomolecules with varying dielectric properties and Reduction in the tunneling barrier height, facilitating improved charge transport.

The drain current variations (Figure 15) reveal that the proposed InP/AlGaAs VTFET achieves a peak sensitivity of 3.5×10^3, outperforming conventional TFETs due to the twin-gate electrostatic modulation strategy. The drain current variation of the proposed device is shown in Fig.15. The OFF-state current is reduced by 103 times for increase in channel length. However the subthreshold swing turns out to be less steep, which degrades the device performance by short channel effects. The validation of the model is assured with results that matched between analytical model and simulation. The impact of drain current variation for different set of work functions shows that drain current is very sensitive to the channel thickness of the proposed device. The lower gate work function (ϕM1 = 4.4 eV) exhibits lower value of drain current due to lower resistivity of gate. It is also observed that higher gate work function shows better ID in short channel device. The impact of drain characteristics of the developed model depicts that subthreshold current in thick gate oxide seems higher compared to that of thin gate oxide. Tunneling of carriers from valence band of the source to the conduction band of the channel is more by incorporating a

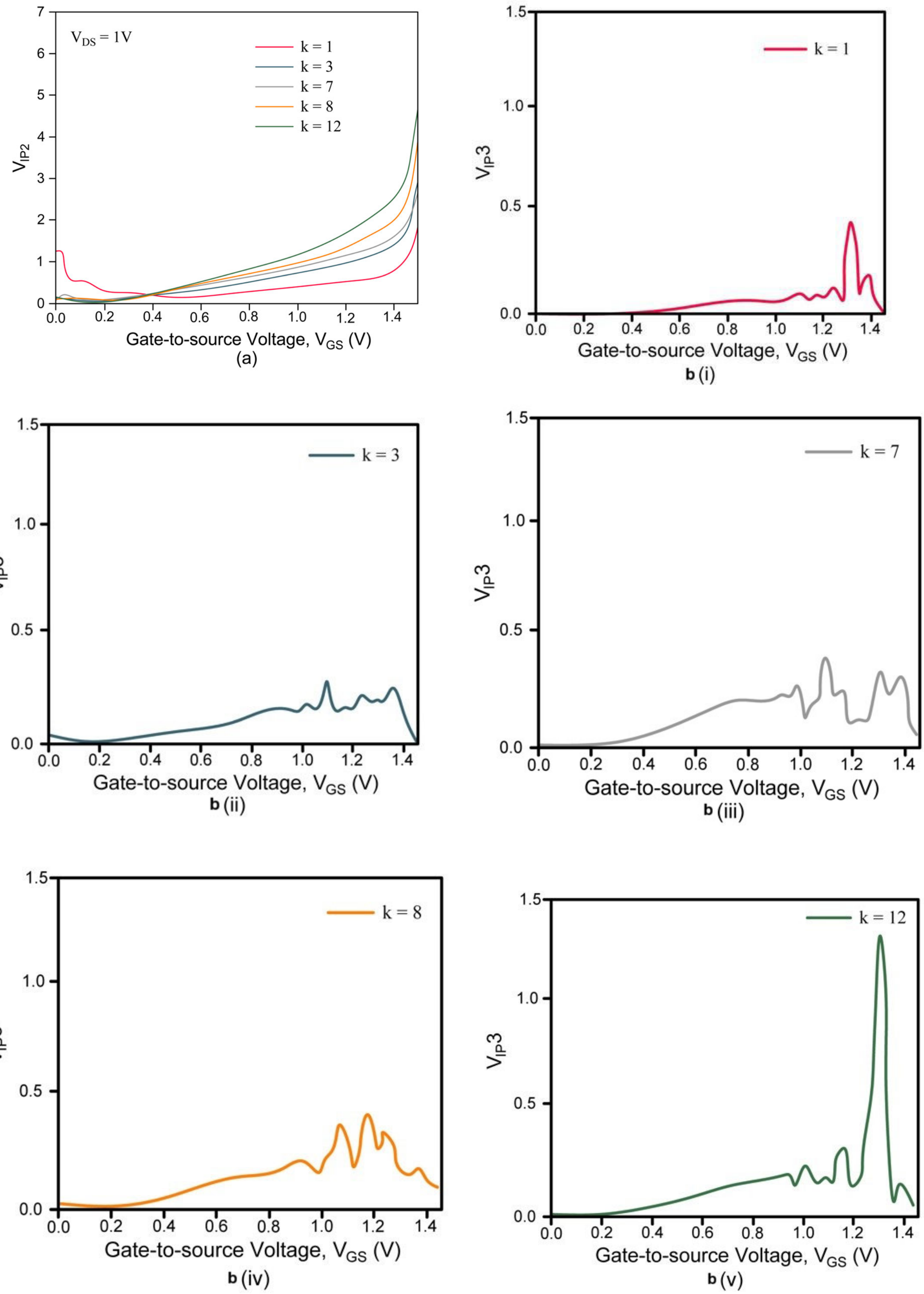

FIGURE 13. Variation of (a) VIP2, (b)VIP3, (c) IIP3 and (d) IMD3 with constant VDS = 0.5 V for various neutral biomolecules.

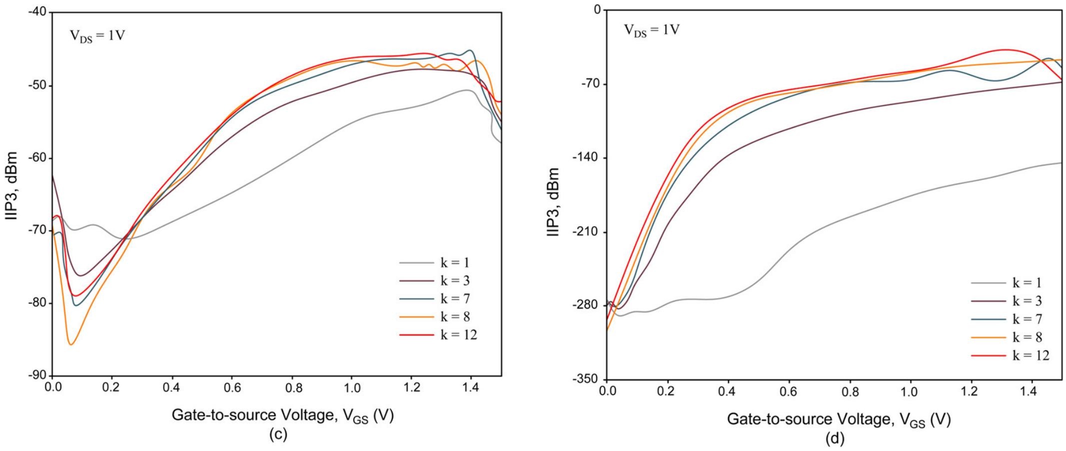

FIGURE 13. *(Continued.)* Variation of (a) VIP2, (b)VIP3, (c) IIP3 and (d) IMD3 with constant VDS = 0.5 V for various neutral biomolecules.

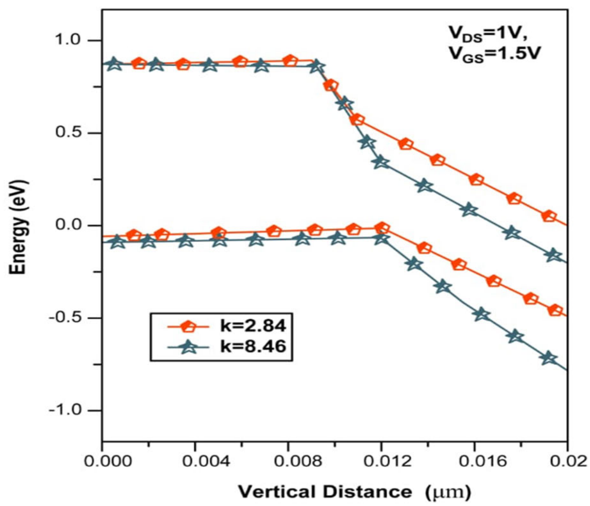

FIGURE 14. Energy band diagram.

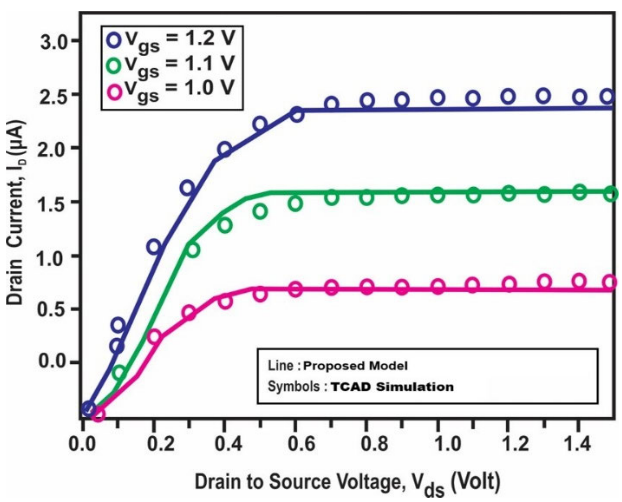

FIGURE 15. Drain Characteristics for different values of Vgs = 1.0,1.1,1.2 V.

higher work function material, we also have low-k material to suppress the TFET's ambipolar behaviour. This proves that the proposed device will be the better dielectric material to combat with short channel effects of the device. Table 3 presents a comparative analysis of the proposed InP/AlGaAs VTFET against conventional TFET biosensors. The electrostatic enhancements in our design result in: Lower SS (due to improved gate control and charge plasma effect), Higher drain sensitivity (due to increased BTBT rate) and Greater scalability for biosensing applications. By integrating charge plasma engineering, optimized nanocavity placement, and auxiliary gate control, the proposed device demonstrates superior electrostatic properties for label-free biomolecule detection.

G. IMPACT OF NANOCAVITY THICKNESS ON ELECTROSTATIC BEHAVIOR AND SENSITIVITY

The nanocavity thickness is a critical parameter influencing the electrostatic properties and sensitivity of the proposed InP/AlGaAs charge plasma dielectric-modulated VTFET biosensor. Since the nanocavity acts as the primary detection region where biomolecules interact with the dielectric medium, its dimensions significantly affect charge modulation, electric field intensity, and drain current characteristics. A reduced nanocavity thickness strengthens gate-to-biomolecule coupling, thereby increasing band-to-band tunneling (BTBT) efficiency and enhancing drain current sensitivity. However, an excessively thin nanocavity can introduce short-channel effects (SCEs) and lead to higher subthreshold leakage currents, which may negatively impact the device's stability. In contrast, a thicker nanocavity diminishes the gate's control over biomolecules, resulting in lower drain current sensitivity but improving device reliability by suppressing leakage currents. Simulation results indicate that nanocavity thickness strongly influences drain current variations, with an optimal range of 4–6 nm providing the best balance between sensitivity and device stability. A nanocavity thinner than 2 nm enhances dielectric modulation but also contributes to higher threshold voltage (V_{th}) fluctuations. Conversely, a thickness exceeding

TABLE 3. Comparison of proposed InP/AlGaAs VTFET with existing TFET configurations.

Parameter	Proposed InP/AlGaAs VTFET (This Work)	Existing Dielectric-Modulated TFET [11]	Conventional TFET [12]
Device Structure	Vertical TFET with charge plasma dielectric modulation	Standard dielectric-modulated planar TFET	Planar TFET without dielectric modulation
Metal Gate Material	TaN (Optimized for enhanced performance)	TiN or other conventional metals	TiN or Mo-based metal gates
Biosensing Mechanism	Nanocavity beneath gate metal near tunneling junction	Surface-based dielectric modulation	No dedicated biosensing mechanism
Peak Drain Current Sensitivity (cm^{-2})	2.5×10^9	Lower than proposed	Significantly lower
Maximum Sensitivity	3.5×10^3	$1.5 - 2 \times 10^3$	$< 1 \times 10^3$
Detection of Biomolecules	Highly sensitive (Detects biomolecules with varying dielectric constants and charge distributions)	Moderate sensitivity	Limited detection capability
Short-Channel Effects (SCEs) Mitigation	Strong (Twin metal gate with lateral dielectric split)	Moderate	Weak (Suffers from high SCEs)
Energy Band Diagram Optimization	Optimized via source-pocket overlap	Limited optimization	Unoptimized
Subthreshold Swing (SS)	Lower SS (Improved due to auxiliary gate and source-pocket overlap)	Moderate	Higher SS
Drain Current Variation	Significant due to charge plasma modulation	Lower than proposed	Minimal
Electrical Optimization	Enhanced via auxiliary gate	No auxiliary gate	No auxiliary gate
Scalability for High-Sensitivity Detection	High	Moderate	Low
Tunneling Efficiency	Improved due to source-pocket overlap	Weaker	Weak (Higher barrier)

6 nm reduces tunneling efficiency, impairing the sensor's ability to distinguish biomolecules with varying dielectric constants. Variations in electric field distribution caused by changes in nanocavity thickness alter charge plasma formation at the tunneling junction, thereby affecting subthreshold swing (SS) and overall device performance.

For an efficient and scalable biosensing platform, nanocavity thickness should be optimized based on biomolecule size and required detection sensitivity. A thinner nanocavity 94–4 nm) combined with high-k dielectrics (HfO2, Al2O3) is ideal for high-sensitivity biosensing, whereas a moderate thickness (4–6 nm) ensures stability and reproducibility in practical applications. Additionally, incorporating a twin-gate architecture and charge plasma modulation techniques helps counteract SCEs in ultra-thin nanocavities, thereby maintaining a high signal-to-noise ratio for biomolecule detection. These results underscore the importance of precise nanocavity engineering in optimizing the performance of InP/AlGaAs VTFET-based biosensors.

IV. CONCLUSION

The InP/AlGaAs Vertical TFET (VTFET) biosensor has been explored for label-free biomolecule detection using dielectric modulation in a cavity under the tunnel gate. This study evaluates its electrostatic properties, including energy band diagrams, drain characteristics, sensitivity, subthreshold swing, and RF performance. The sensor's response is highly dependent on the cavity length. Shorter cavities ($\sim$5 nm) require higher gate voltages to reach optimal drain currents, while an ideal cavity length of 10 nm achieves peak drain sensitivity. Sensitivities of 1.5×10 and 2.5×10 were achieved for positively charged and neutral biomolecules (dielectric constant ($k = 7$), respectively. The sensitivity is notably influenced by both biomolecule charge and cavity dimensions, increasing with longer cavities. In terms of RF performance, transconductance (gm) and gate-to-drain capacitance increase with gate voltage VGS > 0.5 V) and (k), while the cutoff frequency (fT) and gain-bandwidth product rise almost linearly, indicating high-frequency potential. With robust electrostatic control, enhanced sensitivity, and favorable RF metrics, the InP/AlGaAs-VTFET biosensor stands as a promising tool for high-sensitivity, label-free biomolecule detection in advanced biosensing applications.

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